Multilabel Sample Augmentation-Based Hyperspectral Image Classification

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***Abstract*— The quantity and quality of training samples have a great influence on the performance of most hyperspectral image classification approaches. However, in a real scenario, manually annotating a large number of accurate training samples is extremely labor-intensive and time-consuming. In this article, a multilabel training sample augmentation method is proposed. Instead of giving an exact label to each pixel, we just precisely label a small number of pixels by giving them a single label (called single-label samples) and annotate a large number of pixels in certain regions together by giving them multiple labels (called multilabel samples). Furthermore, in order to make full use of the multilabel training samples, a superpixel segmentation and recursive filtering-based method is proposed. The proposed method consists of the following major steps: recursive filtering-based feature extraction, superpixel-based segmentation, and spectral–spatial similarity-based mislabeled sample removal. Experimental results demonstrate that the proposed method can significantly improve the classification accuracy of multiple classifiers by using the multilabel training samples.**

***Index Terms*— Classifier, hyperspectral image classification, multilabel samples, sample augmentation.**

I. Introduction

THE fast-growing trend of hyperspectral remote sensor technology allows the simultaneous acquisition of hundreds of spectral wavelengths for each image pixel. Due to this advantage, hyperspectral images have been used in a variety of remote sensing researches including classification [1], object detection [2], [3], spectral unmixing [4], [5], and so on. Hyperspectral image classification has become an active research topic in the field of aerial and satellite image analysis, which aims at distinguishing different objects of interest in the scene.

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For most of the existing classification methods, by incorporating the label information of training samples, powerful machine learning algorithms are employed to train a decision rule to predict the labels of the testing pixels [6], [7]. A simple solution to improve the classification accuracy is that we can label adequate samples for training. However, in real-world applications, labeled pixels are usually very limited because labeling large amounts of pixels is very time-consuming and requires considerable expertise. When the number of training samples is limited, the classification performance of some hyperspectral classification methods decreases significantly.

The multilabel problem, which is a general and practical problem, has been introduced in the field of computer vision. Multilabel samples and multilabel classification are popular and prevalent in many real-world applications where data instances may be associated with multiple labels. Multilabel samples mean assigning to each sample a set of target labels, which can be thought as predicting properties of a data-point that are not mutually exclusive. For example, Sun *et al.* [8] propose a coevolutionary multilabel hypernetwork to exploit label correlations, which exploits arbitrary order label correlations effectively and has linear computational complexity with respect to the number of labels. In [9], a flexible deep convolutional neural network (CNN) framework, namely, hypotheses-CNN-pooling (HCP), is proposed for multilabel image classification, which is robust to noisy and redundant hypotheses. Luo *et al.* [10] propose a multiview matrix completion (MVMC) framework to fuse different kinds of features for transductive multilabel image classification, which can explore the complementary properties of different views. These multilabel classification methods are originally designed for exploiting label correlations and classification of the data whose multiple labels characterize the multiple semantic meanings they have simultaneously. It is different from binary or multiclass image classification in which only one label can be assigned to each sample.

Multilabel image classification also has been introduced in the field of remote sensing because most remote sensing images contain objects belonging to multiple different categories. For example, Karalas *et al.* [11] introduce the application of multilabel for land cover classification of remote sensing images. The multilabel of a pixel is given by integrating different spatial resolution remote sensing data. In [12], a novel end-to-end network is proposed for multilabel classification of the high-resolution aerial image, which consists of a feature extraction module, a class attention learning layer,

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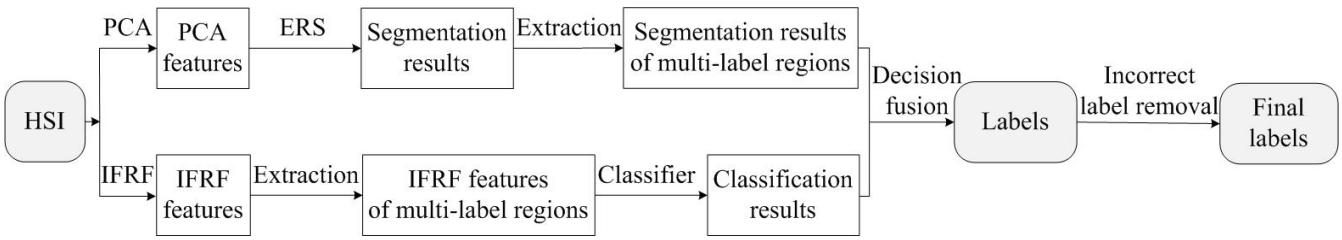


Fig. 1. Schematic of the proposed method.

and a bidirectional long short-term memory (LSTM)-based subnetwork. Wang *et al.* [13] propose a multilabel hyperspectral image classification method based on deep learning, which can exploit the multilabel information that inherently exists in the mixed pixels. In this study, multiple labels are assigned to each pixel since the mixed pixels commonly exist in the hyperspectral image. Although some preliminary research has been done on multilabel remote sensing image classification, it is still an urgent technical problem. Based on the abovementioned survey of related work, each sample is annotated with multiple labels, which represent the classes of objects included in the sample.

Many supervised classification methods have shown their ability to enhance the classification performance. These methods are designed with a single-label pattern, and the performance of most of these methods relies on the number of initially labeled training samples. In the field of hyperspectral remote sensing, collecting training samples is a critical problem since it has a direct impact on most of the subsequent image processing and analysis steps. Generally, for most of these studies in the field of hyperspectral remote sensing, after the data acquisition process, we need to artificially label some samples in the data at the pixel level. The training samples are labeled by the following two main approaches: 1) *in situ* observation and 2) photo interpretation [14], which depends on the expertise of the human users. The manual labeling process is actually labor-intensive and time-consuming. In real-time scenarios, one of the most important problems is the scarcity of available labeled training samples due to complexities and costs that characterize the process of annotation. Accordingly, it may lead to weak estimates of the classifier parameters and potentially high classification error rates because the classification process is constrained to be carried out with limited numbers of training samples.

We have drawn our inspiration from this: if samples can be simultaneously characterized by multiple labels, we can annotate pixels in several regions together by giving these pixels multiple labels as weak label information. Annotating a region containing a large number of pixels means that class labels of all pixels in a region are simultaneously assigned to each pixel, which can reduce the pressure of human annotation compared to single-label annotation. The region to be labeled contains more than one class of object, and thus each pixel possesses multiple labels rather than one label. Therefore, we extend and develop the concept of multilabel to the field of training sample annotation problems in hyperspectral image classification. The multilabel sample augmentation strategy is proposed to label the samples at the region level and reduce the number of single-label samples required to be labeled. During the process of multilabel sample annotation according to visual interpretation, we do not need to distinguish the exact label of each pixel, and a large number of pixels can be labeled together by giving the same multilabels. Instead of labeling a large number of single-label samples, we just need to label a small number of accurate single-label samples and annotate multilabels to a large number of pixels in regions together. To avoid confusion, in this article, "single-label training samples" refer to those pixels with one exact label while "multilabel training samples" refer to those pixels annotated by region which originally have multiple labels. Compared to the expensive labels, the proposed method can reduce the pressure of human annotation and provide weak label information for classification.

In this article, focusing on collecting training samples with human intervention, we first formulate multilabels for each pixel for assisting a user from labeling training samples. Then we execute classification based on pre-setting label-pattern by developing a novel multilabel sample augmentation method, which can make full use of the multilabel training samples. The proposed method consists of the following major steps: first, the image fusion and recursive filtering (IFRF) method is used to extract the spectral–spatial features of the hyperspectral images. Then, an oversegmentation algorithm is adopted to cluster the hyperspectral image into many superpixels. For each multilabel region, the corresponding IFRF features and segmentation results are extracted. Next, different classifiers are used to classify samples in each multilabel region with single-label samples belonging to the corresponding classes, and the classification results are fused with the segmentation results. The aim is to refine the preliminary classification results with the segmentation results, thus improving the label accuracy. In this way, the multilabel samples can be converted to single-label samples. Finally, we remove the inaccurate classified multilabel samples based on the spatial and spectral similarity between the single-label and classified multilabel samples. The proposed method is illustrated in Fig. 1.

The multilabel annotation method can randomly select the regions of interest that contain not only a type of object and then label them together, so the experts do not need to distinguish the edges of adjacent objects and accurately label each pixel in the complex scenes. Experiments performed on three real data sets demonstrate the improvement of classification performance related to the same classifiers by using the proposed multilabel sample augmentation method. The proposed

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method can achieve competitive classification accuracy and computational efficiency by using the multilabel samples and can significantly reduce the number of single-label samples that need to be accurately labeled by experts. It means that the proposed method will be quite suitable in real applications, especially for the scenes that only know which classes are contained in the scenes but are difficult to accurately identify their specific locations by photo interpretation.

To sum up, the main contributions of this article are threefold.

1) This article extends and develops the concept of multilabel to the field of training sample annotation for hyperspectral image classification. Especially, different from the previous research on classification with the single-label and multilabel samples, a large number of pixels in certain regions are labeled together by giving them multiple labels, which can reduce the pressure of human annotation and provide weak label information for classification.
2) In order to make full use of the multilabel training samples, a preliminary IFRF feature-based classification result and a superpixel-based segmentation result are combined to convert the multilabels to single-labels. A spectral–spatial similarity measurement is specially designed to remove the mislabeled samples, which can directly quantify how much a pixel belongs to a class. Apart from the spectral–spatial information, it makes use of the multilabel information since the measurement of each multilabel sample is calculated between it and the single-label samples belonging to the same classes. In this way, mislabels generated by the preliminary results have a chance to be deleted. Furthermore, this spectral–spatial similarity measurement can be well adjusted to different applications since its formulation is flexible. Most forms of spectral or spatial similarity measurement can be combined with it. Experimental results show that the proposed method can significantly improve the classification performance of several classical classifiers with the multilabel training samples.
3) By analyzing the limitations of the proposed method, the main challenges of hyperspectral image classification with multilabel training samples are discussed. Since multilabel samples for classification is quite new in the field of hyperspectral image processing and analysis, some new outlooks to the current development are given.

The rest of this article is organized as follows. In Section II, the domain transform recursive filter (DTRF) and the entropy rate superpixel (ERS) segmentation algorithm are reviewed. Section III introduces the details of the proposed multilabel sample augmentation method. Section IV presents and analyzes the experimental results. Conclusions are drawn in Section V. Finally, Section VI gives the outlooks to provide followers with the guidance in this field.

II. Related Works

A. Domain Transform Recursive Filter

Many feature extraction and reduction methods have been proposed for hyperspectral image classification, such as principal component analysis (PCA) [15], linear discriminant analysis (LDA) [16], and extended morphological attribute profiles (EMAP) [17], [18], which can be adopted in our proposed method. However, these transform-based methods may change the physical meaning of pixels.

A recently proposed DTRF has been demonstrated to be quite effective, and has been successfully applied in the field of hyperspectral image processing [19], [20]. First, given an input signal I, an approximation is achieved by calculating the sum of the spatial and intensity differences between two adjacent pixels. The approximate distance-preserving transformation is calculated by the DTRF as follows:

$$U_i = I_0 + \sum_{j=1}^{i} \left(1 + \frac{\delta_s}{\delta_r}|I_j - I_{j-1}|\right) \tag{1}$$

where U_i refers to the domain transformed signal, and δ_s and δ_r are the spatial and range parameters that can control the contribution of smoothness of the filter, respectively.

Then, the 1-D input signal I is processed by recursive filtering as follows:

$$J_i = (1 - a^b)I_i + a^b J_{i-1} \tag{2}$$

where J_i is the filtered result of the ith pixel. a is a feedback coefficient which can be obtained by $\exp(-\sqrt{2}/\delta_s) \in [0, 1]$. b refers to the distance between two adjacent domain transformed samples U_i and U_{i-1}. It can be observed from (2) that a^b tends to be 0 as b increases. Therefore, the propagation chain will be stopped, and we can preserve the sharp edges while pixels on the same side of the edges will tend to obtain similar filtering outputs.

The abovementioned 1-D operations will be iteratively performed by the DTRF along each dimension of the input image. In [21], it has been demonstrated that more than three iterations are able to yield satisfactory filtering results. In this work, we refer the DTRF operation as $\texttt{DTRF}(I, \delta_s, \delta_r)$. I represents the input, δ_s and δ_r are the parameters that control the amount of smoothness. In the original work [21], a complete description of the DTRF can be found.

B. Entropy Rate Superpixel Segmentation

Superpixel segmentation is an important module in the field of computer vision, which can provide efficient representation to facilitate visual recognition [22]–[24]. The shape and size of each superpixel can be adaptively changed according to different spatial structures. Each superpixel is a perceptually meaningful region and is expected to contain objects belonging to only one class.

The ERS segmentation is studied as an undirected graph-based clustering problem [25]. Given a base image and the superpixel number L, a graph $G = (V, E)$ is constructed on the base image firstly. V is the vertex set corresponding to pixels of the base image, and E is the edge set representing the pairwise similarities between adjacent pixels. Then, the graph is segmented into L connected subgraphs by selecting a subset of edges $A \subseteq E$, and each subgraph corresponds to a superpixel. To form the compact, homogeneous, and balanced clusters, the clustering objective function is composed of two

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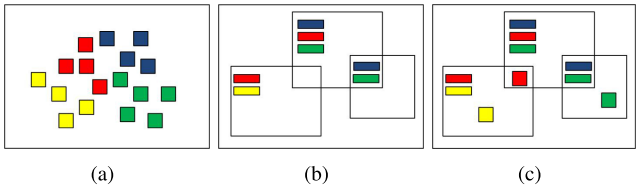


Fig. 2. Training samples with single label or multiple labels. (a) Single-label training samples. (b) Multilabel training samples. (c) Single-label and multilabel training samples. Different colors refer to different classes.

terms: the entropy rate of a random walk on a graph $H(\cdot)$ and a balancing term on the cluster distribution $B(\cdot)$. We can achieve the clustering by optimizing the objective function with regard to the edge set as follows:

$$\max_{A}\{H(A)+\lambda B(A)\} \qquad \text{subject to } A \subseteq E \tag{3}$$

where $\lambda \geq 0$ is the weight of balancing term, which is used to control the contribution of the entropy rate term and balancing term. In fact, the optimization of the objective function in (3) can be efficiently solved by a greedy optimization scheme, and the details can be found in [26]. Moreover, compared to the pixel representation, superpixel representation greatly reduces the number of image primitives and, thus, has high computational efficiency.

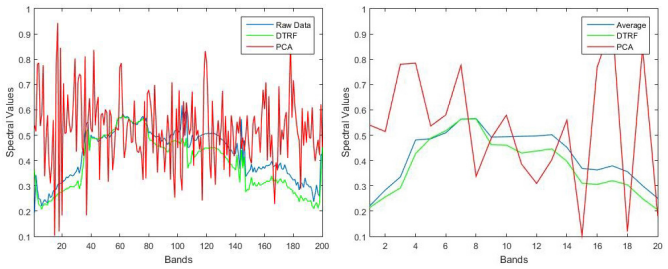
III. Proposed Method

A. Multilabel Sample Augmentation

It is a challenging task to obtain an abundant description of land objects since the high cost and labor-intensive process of human annotation through field studies. Here, we extend the concept of multilabel in computer vision to the training samples annotation problem in the field hyperspectral image classification, which can reduce the pressure of human annotation. Instead of labeling pixels one by one, we annotate a large number of pixels belonging to several certain regions in the scene together [see Fig. 2(b)]. Each region to be labeled contains more than one class of object, and all classes included in this region are assigned to each pixel. Therefore, each pixel possesses multiple labels rather than one label, which called multilabel training samples in the article. The single-label samples annotation is costly but can provide a large amount of exact label information. The multilabel samples annotation is inexpensive but provides less information. It is a viable strategy to annotate a small number of single-label samples and a large number of multilabel samples for classification [see Fig. 2(c)]. A small quantity of single-label samples can provide precise label information and multilabel samples can provide weak label information for training. In this way, the requirements of manpower and resources can be efficiently reduced.

Here, sample augmentation has two meanings.

1) *The Augmentation of Multilabel Samples:* A multilabel sample annotation method is proposed to reduce the cost of labeling and improve the efficiency of labeling. In this way, a large number of multilabel training samples that provide weak label information can be obtained.



(a) (b)

Fig. 3. Original spectra of a randomly selected pixel and its corresponding spectral values of the features obtained by the average, DTRF and PCA methods. (a) Original spectra of this pixel and the spectral values of this pixel with 200 bands after the DTRF and PCA operations. (b) Spectral values of this pixel with 20 bands after the average, DTRF, and PCA operations.

2) *The Augmentation of Input Single-Label Training Samples for Classification:* A classification method for multilabel regions in the hyperspectral images is proposed. It refers to the conversion of multilabel samples to single-label samples, which can increase the number of single-label training samples that can be directly used for classification. Most of the classification methods have shown their ability to enhance the classification performance with single-label training samples, if the multilabel training samples can be converted to single-label samples, they can be directly used to train the classifiers.

B. IFRF-Based Feature Extraction

In the proposed sample augmentation method, a realtime edge-preserving filtering method, i.e., the DTRF is adopted, since it is quite effective for feature extraction and can improve the performance of hyperspectral image classification [19]. Fig. 3 shows the original spectra of a randomly selected pixel and its corresponding spectral values of the features obtained by the DTRF and PCA methods. Fig. 3(a) shows the original spectra of this pixel and the spectral values of the pixel with 200 bands after the DTRF and PCA operations. Fig. 3(b) shows the spectral values of this pixel with 20 bands after the average, DTRF, and PCA operations. As Fig. 3 shows, the IFRF features can well preserve the physical meaning of the hyperspectral data and the resulting feature images still represent the spectral values in a specific spectral range. From a visual analysis, the features appear to be spectrally consistent with respect to the original hyperspectral image. The IFRF method is used to extract the spectral–spatial feature of a hyperspectral image as follows.

Given a hyperspectral image $\mathbf{H}$, the IFRF-based feature extraction method is performed: First, since the adjacent bands of the hyperspectral image usually contain redundant information and they look quite similar, the original hyperspectral image $\mathbf{H}$ is first segmented into K subgroups of equal size along the spectral dimension, in which K is the number of subgroups. Then, the averaging band spectral values of each subgroup are calculated, and the dimension of the hyperspectral image can be reduced to K. Finally, we can obtain the

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IFRF features of the dimension-reduced hyperspectral image by using the DTRF as follows:

$$\mathbf{F_K} = \mathtt{DTRF}(\mathbf{H}_{\mathrm{K}}, \delta_{\mathrm{s}}, \delta_{\mathrm{r}}) \tag{4}$$

where $\mathbf{H}_K$ refers to the dimension-reduced data with K bands, DTRF refers to the domain transform recursive filtering operation, the parameters δ_s and δ_r are the parameters that can control the amount of smoothness, respectively. If better and more efficient feature extraction methods were adopted, the outcomes of most classifiers or classification methods, including ours, could be better.

C. Superpixels-Based Segmentation

Since the hyperspectral image usually has hundreds of spectral bands, the PCA [15] is first used to reduce the spectral dimension of the hyperspectral image $\mathbf{H}$, which can also improve the computational efficiency. Most spectral information is retained in the first three principle components, and thus we use them as the base images. Then, the oversegmentation algorithm ERS [25] is applied on the dimension-reduced image $\mathbf{H}_p$ to generate a 2-D superpixel map

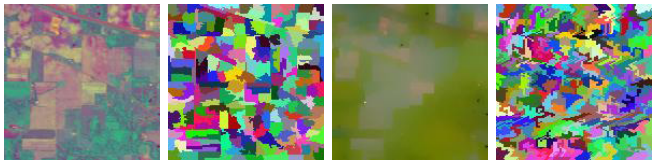
$$\mathbf{H}_p = \bigcup_{k}^{L} \chi_k, \quad \text{s.t. } \chi_k \cap \chi_g = \emptyset \tag{5}$$

where L denotes the number of superpixels, χ_k is the kth superpixel. The setting of L can be selected based on the complexness of the spatial texture in the hyperspectral images. In order to exploit the texture information of different hyperspectral images, a simple texture detector, i.e., the Laplacian of Gaussian (LoG) [27], is applied on the base images. The texture complexity of hyperspectral images can be measured by comparing the number of nonzero elements in the detected edge image with the total number of pixels in the base image. If the texture complexness of hyperspectral images is higher, the number of superpixels is larger, and thus L is selected adaptively as follows:

$$L = L_{\text{base}} \frac{N_F}{N_I} \tag{6}$$

where N_F is the number of nonzero elements in the filtered images, N_I denotes the total number of pixels in the base image I_f, L_{base} is a predefined number. It should be mentioned that if more advanced texture detectors are used, the performance will be enhanced, but the computational cost may be increased.

Especially, Fig. 4 shows the segmentation results obtained by using the IFRF and PCA features as the base images. It can be seen from Fig. 4(c) that although the boundaries of the image are preserved, the averaging and smoothing operations may lose much information and decrease the differences among pixels belonging to different classes. We adopt the first three principle components of the hyperspectral images as the base images since most information can be retained in them, which can not only reduce the dimension of the data but also increase the spectral separability of pixels [28]. As Fig. 4(b) and (d) shows, the segmentation result obtained based on the PCA features is better than the result obtained by using the DTRF features as the base image. Furthermore, another advantage of the PCA is that it is an unsupervised method, and thus, there is no need to manually select the parameters to obtain a satisfiable result, which increases the operational applicability in real applications. It should be noted that the feature extraction process in the proposed method is optional, and other state-of-the-art methods can also be equally used.



(a) (b) (c) (d)

Fig. 4. Segmentation results obtained by using different base images. (a) PCA features. (b) Segmentation result obtained by using the PCA features as the base images. (c) IFRF features. (d) Segmentation result obtained by using the IFRF features as the base images.

After obtaining this 2-D superpixel map, we can get the position indexes of pixels within each superpixel. The position indexes for L superpixels in the 2-D map can be applied to the IFRF feature maps obtained earlier, and thus, the corresponding L nonoverlapping 3-D superpixels of the feature maps can be extracted. In this way, the superpixels in IFRF features can be obtained. Next, different classifiers are used to classify pixels in each multilabel region with corresponding single-label training samples whose labels are included in the multilabel. Each superpixel is expected to contain objects belonging to only one class. For multilabel samples classification results, the labels of pixels in each superpixel are corrected to the class with the largest number in this superpixel. In this way, the multilabel samples could be roughly converted to single-label samples.

The ERS method is chosen for segmentation since it performs quick oversegmentation for hyperspectral images. The ERS is an alternative segmentation method to be combined in the proposed method. If highly accurate segmentation methods were adopted, the outcomes of our method could be certainly improved. However, it would be difficult to tell whether the improvement of final classification results should be credited to the feature extraction/segmentation techniques or the proposed multilabel samples method.

D. Spectral–Spatial Similarity-Based Mislabeled Sample Removal

In order to measure the spectral–spatial similarity between the single-label samples and the classified multilabel samples, the spectral and spatial distances between them can be represented by calculating Euclidean distance (ED), spectral angle metric (SAM), spectral correlation coefficient (CC), and so on. Let A_{ci} denote the spectral–spatial similarity between the classified multilabel sample x_i and the single-label samples belonging to class c, which means how much x_i belongs to c. If the value of A_{ci} is larger, the spectral–spatial similarity is

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stronger. A_{ci} is formulated as follows:

$$A_{ic} = \frac{\left(\sum_{j \in \Omega_n^c} s_{ij}\right)/n}{\sum_{c=1}^{C} \left(\sum_{j \in \Omega_n^c} s_{ij}\right)/n} \tag{7}$$

where n is the number of single-label samples belonging to c, Ω_n^c refers to the set of indexes of the single-label samples belonging to c, and C represents the number of class. Especially, in order to make sure that A_{ci} ranges from 0 to 1, the denominator is used for normalization. Here, spectral CC and spatial ED are used to measure the spectral–spatial similarity, and s_{ij} can be calculated by

$$s_{ij} = \exp(0.5 \cdot (\rho_{ij} - d_{ij})) \tag{8}$$

$$\rho_{ij} = \frac{\mathtt{Cov}(x_i, x_j)}{\sqrt{\mathtt{D}(x_i)\mathtt{D}(x_j)}} \tag{9}$$

where ρ_{ij} refers to the CC between samples x_i and x_j, d_{ij} refers to the spatial ED between samples x_i and x_j, and Cov and D refer to the covariance and the variance operations, respectively.

For each multilabel sample x_i, if the corresponding c of maximum value A_{ci} is the same as the single-label obtained, this sample is retained, otherwise, this sample is deleted. Finally, based on different efficient feature extraction methods, both the single-label and the remained multilabel samples can be used as training samples for the classifier to obtain the final classification results.

For each classified multilabel sample, the similarity between it and the single-label samples belonging to the multiple labels (not only the same class but also the different classes) is measured, and thus, the similarity between the same and different classes has been considered. Although the difference between the same class is not taken into account, the proposed method can remove the mislabeled samples to some extent because of the constraints on spatial information. If the difference between the same class and the similarity between different classes are taken into account, the outcomes of our method could be certainly improved. It will serve as our future work to further optimize this work.

The proposed method can be regarded as a decision fusion-based classification operation for multilabel regions, which aims at converting the multilabels of samples to single-labels, thus obtaining single-label sample argumentation. For Section III-B–III-C, the preliminary classification result of the multilabel region is refined with the segmentation result, thus improving the accuracy of obtained single-labels. In this way, the multilabel samples can be converted to single-label samples. Section III-D can be regarded as a post-processing step, we can remove incorrect classified multilabel samples based on the spatial and spectral similarity measurement between the single-label and classified multilabel training samples.

E. Details of the Proposed Algorithm

For a detailed illustration of the proposed method, the pseudocode of the proposed algorithm is illustrated in Algorithm 1. Before depicting this algorithm, the used notation is first introduced. Let $\mathbf{H}$ represent the input hyperspectral image. Let $D = (x_i, Y_i)_{i=1}^{|D|}, Y_i = \{y_1, \ldots, y_m\}$ be the multilabel data set, in which x_i refers to the ith multilabel sample, Y_i is its multilabel, m is the number of labels, and $|D|$ is the number of multilabel samples. Let $S = (s_i, z_i)_{i=1}^{|S|}$ represent the single-label data set, in which s_i refers to the ith single-label sample, z_i is its single-label, and $|S|$ is the number of single-label samples. Let $S_m = (s_i, z_i)_{i=1}^{|S_m|}, s.t. z_i \in Y_i$ represent the single-label data set consisting of the single-label

Algorithm 1 Algorithm of the Proposed Method

Input:

Hyperspectral image, $\mathbf{H}$;

Multilabel training set, $D = (x_i, Y_i)_{i=1}^{|D|}, Y_i = \{y_1, \ldots, y_m\}$;

Single-label training set, $S = (s_i, z_i)_{i=1}^{|S|}$.

Output:

Final classified multilabel training set, $D_f = (x_i, t_i)_{i=1}^{|D_f|}$.

Reduce the dimension of $\mathbf{H}$ based on the averaging method to obtain the dimension-reduced image $\mathbf{H}_K$.
According to 4, obtain the IFRF features of the image, i.e., $\mathbf{F}_K$.
Reduce the dimension of $\mathbf{H}$ based on the PCA method to obtain the dimension-reduced image $\mathbf{H}_p$.
According to 5, obtain the segmentation map, i.e., $\bigcup_k^L \chi_k, s.t. \chi_k \cap \chi_g = \emptyset$.
Extract the corresponding IFRF features $\mathbf{F}_{Km}$ and segmentation results of multilabel regions.
for each $i = 1 : |D|$ **do**
    Classify x_i based on $\mathbf{F}_{Km}$ by using the classifiers trained with the single-label samples $S_m = (s_i, z_i)_{i=1}^{|S_m|}, s.t. z_i \in Y_i$, to obtain the rough label t_i.
end for
Construct the classified multilabel training set $D_c = (x_i, t_i)_{i=1}^{|D_c|}$.
for each $i = 1 : |D_c|$ **do**
    $t_i = t_{is}$, t_{is} is the most numerous class in the superpixel $\chi_k, s.t. x_i \in \chi_k$.
end for
Construct the refined multilabel training set $D_r = (x_i, t_i)_{i=1}^{|D_r|}$.
for each $i = 1 : |D_r|$ **do**
    According to 7, calculate the spectral–spatial similarity measurement A_{ci}, obtain the max A_{ci} and the corresponding c.
    if $t_i = c$ **then**
        This sample x_i is remained.
    else
        This sample x_i is deleted.
    end if
end for
Construct the final multilabel training set $D_f = (x_i, t_i)_{i=1}^{|D_f|}$.
return $D_f = (x_i, t_i)_{i=1}^{|D_f|}$.

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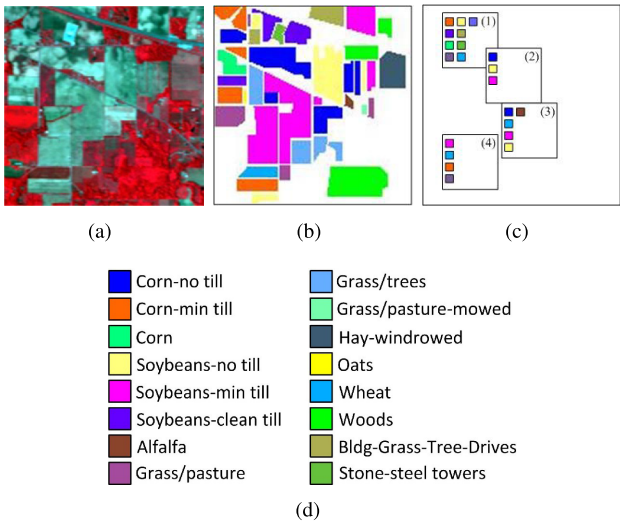


Fig. 5. Indian Pines data set. (a) Three-band color composite of the Indian Pines image. (b) Ground truth. (c) Multilabel training samples. (d) Class names.

samples whose labels are contained in the multilabels. Let $D_f = (x_i, t_i)_{i=1}^{|D_f|}$ be the final classified multilabel training set with single-labels, in which $|D_f|$ refers to the number of remained multilabel samples.

IV. Experiments

A. Data Sets

In the experiments, three hyperspectral data sets are used and the details of them are presented as follows.

The first data set, Indian Pines data set, was acquired by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). The data cover a mixed vegetation site and have a spatial size 145×145. These data contain 220 bands in the wavelength range of 0.5–2.5 μm, and 200 spectral bands are used after discarding 20 water absorption bands (no. 104–108, 150–163, and 220). The false-color image and ground truth are presented in Fig. 5(a) and (b). The correspondence between sixteen different object classes and colors is shown in Fig. 5(d).

The second real data set is Salinas data set, which was acquired by the AVIRIS sensor. This image is a site over the agricultural area of Salinas Valley, California. The data with a spatial resolution of 3.7 m contains 224 bands and has a spatial size of 512×217. In the experiment, only 204 spectral bands are remained for analysis after removing water absorption bands (no. 108–112, 154–167, and 224). The false-color image and ground reference data are presented in Fig. 6(a) and (b). Fig. 6(d) shows the correspondence between sixteen different land-cover classes and different colors.

The third real data set is the Kennedy Space Center (KSC) data set, which was acquired by the AVIRIS over the Kennedy Space Center, Floride. The image has a spatial size of 512×614 and 176 bands after omitting 48 water absorption and low SNR bands. The false-color image and ground truth are presented in Fig. 7(a) and (b). The correspondence between nine different land-cover classes and colors is shown in Fig. 7(d).

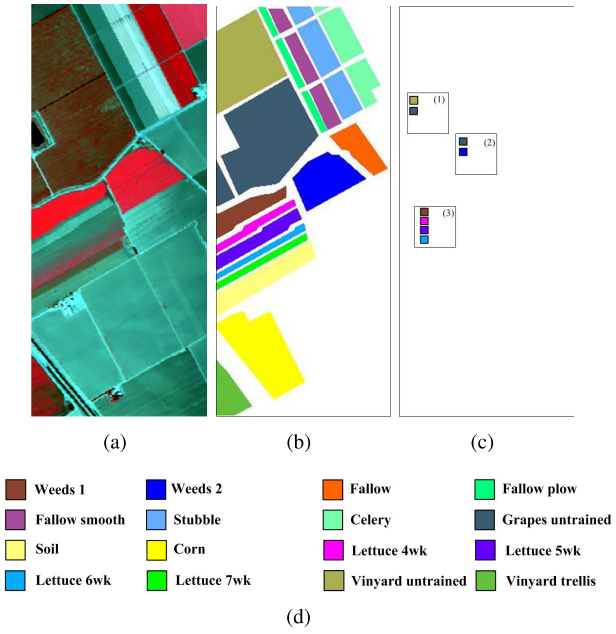


Fig. 6. Salinas data set. (a) Three-band color composite of the Salinas image. (b) Ground truth. (c) Multilabel training samples. (d) Class names.

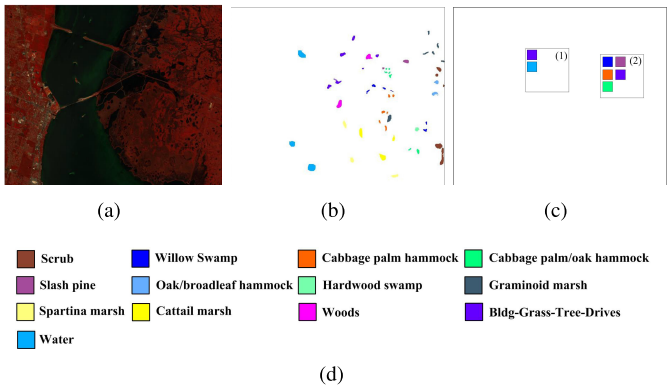


Fig. 7. KSC data set. (a) Three-band color composite of the KSC image. (b) Ground truth. (c) Multilabel training samples. (d) Class names.

B. Generation of Multilabel Samples

In this article, we introduce a novel multilabel sample augmentation approach to reduce the burden of human labeling for hyperspectral image classification. However, no ready specific sources of multilabel samples are available for our analysis. In order to construct the multilabel samples of these three hyperspectral data sets, the public ground truths of the data sets, i.e., the single-label matrixes, are divided into several nonoverlapping blocks using a square grid of certain sizes. We select regions containing over two classes by excluding grids composed of only one label. Therefore, we acquire a reasonable scenario for the multilabel samples. Taking the Indian Pines data set for example, Fig. 5(c) shows the constructed multilabel samples, which contain four multilabel regions. It should be noted that the multilabel of each pixel contains both right single-label and the labels of other pixels in the region. Table VI shows the multilabel of the samples contained in these multilabel regions. It is important

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TABLE I

STATISTICAL CHARACTERISTICS OF THE CONSTRUCTED MULTILABEL SAMPLES OF PUBLICLY AVAILABLE HYPERSPECTRAL DATA SETS. iTH REFERS TO THE iTH MULTILABEL REGION [SEE FIGS. 5(c)–7(c)]. D REFERS TO THE MULTILABEL REGION WHICH CONTAINS $|D|$ MULTILABEL SAMPLES. m IS THE NUMBER OF MULTILABEL

Metrics	Indian Pines		Salinas		KSC	
	$\|D\|$	m	$\|D\|$	m	$\|D\|$	m
1-th	1013	9	2127	2	327	2
2-th	1135	3	2321	2	373	5
3-th	1041	5	1927	4	None	0
4-th	1147	4	None	0	None	0
LC	5.1464		2.6045		3.5985	
LD	0.3217		0.1628		0.2768	

to highlight that not all pixels in the ground truths of public data sets are labeled, and only labeled pixels can provide label information for the multilabel samples. Therefore, in the following experiments, only labeled pixels of the ground truth are remained in the multilabel regions and are given multilabel. The reason is that for hyperspectral image classification methods, we usually assume that the training set is highly convincing.

Furthermore, not all multilabel data sets are equal, even if they have the same number of samples or labels. We estimated certain statistical metrics in Table I to obtain a better understanding of the characteristics of the constructed multilabel data set [29]. In Table I, D refers to the multilabel data set, which contains $|D|$ multilabel examples (x_i, Y_i), $i = 1, \ldots, |D|$. x_i is the ith instance and Y_i is its multilabel. Label cardinality (LC) can be calculated as follows by measuring the average number of class labels associated with each instance:

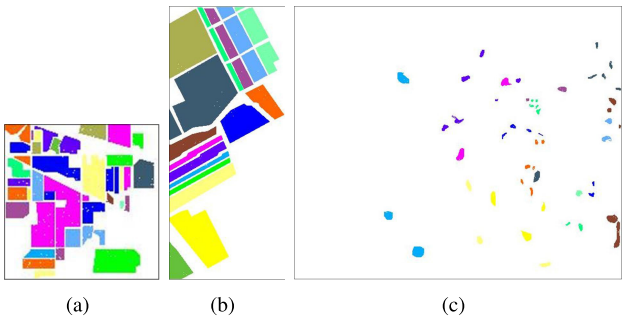
$$\mathrm{LC}(D) = (1/|D|) \sum_{i=1}^{|D|} |Y_i|. \tag{10}$$

LC is the number of possible labels of the $|D|$ examples in a multilabel data set. The larger the value of LC, the more difficult it is to obtain good classification performance.

Another evaluation metric, label density (LD), is also adopted, which can be calculated as follows:

$$\mathrm{LD}(D) = (1/|D|) \sum_{i=1}^{|D|} (|Y_i|/m) \tag{11}$$

where m is the number of labels. LD is the cardinality normalized by m, and thus, quantifies how dense or sparse the multilabel region is. Taking the KSC data set for example, the constructed multilabel samples consist of two regions, we regard them as the 1th and 2th multilabel regions [see Fig. 7(c)]. The 1th multilabel region contains 373 multilabel samples and each sample has 5 labels, the 2th multilabel region contains 327 samples and each sample has 2 labels. The values of LC and LD are 3.59 and 0.27, respectively. It can be seen that the difficulty of multilabel classification of these three data sets in descending order is: the Indian Pines image, the KSC image, and the Salinas image.



(a) (b) (c)

Fig. 8. Test sets used in the experiments. Because of space constraints, we only show the test samples randomly selected in one of the experiments. (a) Test set of Indian Pines data set. (b) Salinas data set. (c) KSC data set.

C. Effect of the Number of Superpixels and Different Features

Here, the effect of the number of the base superpixel L_{base} on the performance of the proposed method is analyzed. For a fair comparison, the reported accuracy values for this analysis experiment are the average results over ten runs with randomly selected single-label training samples. The remaining labeled samples in the ground truths are used for testing. For the three data sets, the test samples randomly selected are presented in Fig. 8. Because of space constraints, we only show the test samples randomly selected in one of the experiments. In each classification experiment, the same single-label training set and test set are used. In this analysis, the numbers of single-label training samples selected from the Indian Pines, Salinas, and KSC images are the same as those shown in Tables III–V.

Fig. 9 presents the overall accuracy (OA), average accuracy (AA), and Kappa coefficient values of the proposed method under different base superpixel numbers on the three hyperspectral images. The base superpixel number is selected from 1400 to 2500. It can be observed that, as the base superpixel number varies from 1500 to 2500, the OA, AA, and Kappa values of the proposed method show good performance on the three hyperspectral images in most cases. When the number of the base superpixel is less than 1500, the OA, AA, and Kappa values of the proposed method drop significantly in a few cases. When the base superpixel number increases from 1500 to 2500, the OA values of the proposed method will be stable on the three images. Therefore, the number of the base superpixel is set as 2000 in our experiments.

Furthermore, in order to verify the generalization ability of the multilabel sample augmentation method for different features, the influences of different features used in the proposed method are also analyzed. An experiment is performed on the Indian Pines data set with the same experimental settings. Table II shows the classification performance of the proposed method with the SVM classifier and several widely used feature extraction methods, i.e., the EMAP [17], [18], the intrinsic image decomposition (IID)-based features [30], [31], the PCA-based edge-preserving features (PCA-EPFs) [20], the IFRF features [19], and the fused segmented PCA-based Gaussian pyramid features (SPCA-GPs) [28]. For the EMAP method, we use four principal components of the hyperspectral

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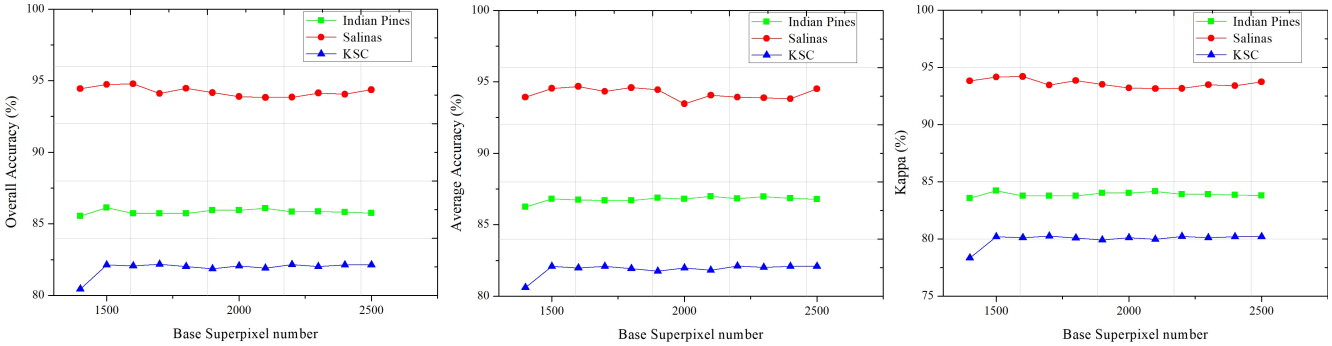


Fig. 9. Effect of the base superpixel number L_{base} when applying the proposed method to the three test images. (a) OAs recorded on the three test images. (b) AAs. (c) Kappas.

TABLE II

OAs, AAs and Kappa Coefficients Obtained by the Proposed Method Using Different Features. The Experiment Is Performed on the Indian Pines Data Set When the Number of Training Samples is the Same as Those Shown in Table III. "S" Refers to Classification With Single-Label Training Samples, "S+M" Refers to Classification With Single-Label and Multilabel Training Samples. The Standard Variances of the Results of Ten Repeated Experiments Are Presented in the Parentheses

Metrics	RawData		EMAP [17]		IID [30]		IFRF [19]		EPF-PCA [20]		SPCA-GPs [28]	
	S	S+M	S	S+M	S	S+M	S	S+M	S	S+M	S	S+M
OA	55.19	**63.18**	68.13	**74.81**	78.63	**82.36**	82.04	**85.96**	89.19	**90.53**	89.59	**90.80**
AA	45.24	**46.85**	64.19	**73.53**	81.90	**85.52**	83.04	**86.81**	86.27	**89.31**	89.91	**91.79**
KAPPA	47.95	**56.86**	64.69	**71.27**	75.70	**79.92**	79.55	**84.01**	87.61	**89.19**	88.08	**89.48**

image to compute the EMAP features. The values of the threshold for the area attribute are 100, 500, 1000, and 5000, for the moment of inertia are 0.2, 0.3, 0.4, 0.5, and for the length of the diagonal of the box bounding the region attribute are 10, 25, 50, 100 [18]. For other features, the default parameters settings given in the corresponding works of literature are adopted. These methods are adopted because they are either highly cited in the hyperspectral remote sensing community or their codes are available.

From Table II, it can be observed that these features used in the proposed method give higher classification accuracies with multilabel training samples than just using single-label training samples in terms of OA, AA, and Kappa coefficient. This experiment further demonstrates that most existing feature extraction methods are effective in the proposed method for multilabel sample augmentation. In addition to the IFRF features used in the proposed method, other highly accurate feature extraction methods can be adopted for hyperspectral image classification with multilabel samples, and the classification performance could be certainly improved. In other words, the IFRF-based feature extraction is only an alternative feature extraction tool to be combined in the method. The multilabel samples are indeed efficient, and thus, they will be quite suitable to improve the classification performance with more advanced features.

D. Effect of Different Classifiers on Classification Performance

To validate the effectiveness of the multilabel sample augmentation method, four widely used classifiers in the field of hyperspectral image calcification are used. They are nearest neighborhood (NN) [32], support vector machine (SVM) [33], [34], random forest (RF) [35], [36], and logistic regression via splitting and augmented Lagrangian method (LORSAL) [37]. Since there are no specific multilabel classifiers for hyperspectral images, we compare the performance of these general classifiers by adapting our method. If these simple classifiers are efficient, more advanced classifiers or classification methods can produce better classification results in the presence of multilabel training samples. The recorded values of OA, AA, Kappa, and class individual accuracies (CA) are calculated by averaging the results obtained in ten repeated experiments. In addition, both the mean and standard deviation values of the accuracies obtained in repeated experiments are reported.

First, we perform the experiments on the Indian Pines data set. The multilabel samples generated earlier are still used and only 1% single-label samples are selected randomly for training. As Table III shows, in terms of the classification accuracies, the proposed method using multilabel training samples can obtain higher values in OA, AA, and Kappa coefficient than just using limited single-label training samples. For all classifiers used in the experiment, the classification performance of most classes can be improved by using the multilabel training samples. It can demonstrate that the proposed multilabel sample augmentation method contributes significantly. It should be mentioned that since objects of class 7 to 9 and 14 are not included in the multilabel training set, the classification accuracies of these classes stay the same or go down to some degree. The main reason for the decline in CA is that after the conversation from

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TABLE III

OBJECTIVE ACCURACY INDEXES OBTAINED ON THE INDIAN PINES DATA SET BY DIFFERENT CLASSIFIERS. "_S" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES, "_S+M" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES AND MULTILABEL TRAINING SAMPLES. THE STANDARD VARIANCES OF THE RESULTS OF TEN REPEATED EXPERIMENTS ARE PRESENTED IN THE PARENTHESES

Class	train	test	Classification accuracies obtained by different classifiers (in %)							
			NN_S	NN_S+M	RF_S	RF_S+M	LORSAL_S	LORSAL_S+M	SVM_S	SVM_S+M
Alfalfa	1	45	69.56(0.19)	**88.00(0.23)**	61.56(0.19)	**71.33(0.31)**	**96.44(0.11)**	96.00(0.08)	89.11(0.13)	**97.78(0.01)**
Corn_N	29	1399	71.95(0.08)	**79.75(0.09)**	67.61(0.12)	**72.15(0.10)**	80.45(0.13)	**83.07(0.13)**	82.88(0.07)	**84.43(0.05)**
Corn_M	17	813	67.63(0.10)	**71.26(0.12)**	**61.92(0.13)**	58.53(0.12)	**74.54(0.18)**	72.80(0.17)	69.30(0.12)	**74.71(0.14)**
Corn	5	232	48.97(0.10)	**67.86(0.10)**	24.95(0.07)	**47.47(0.21)**	71.02(0.19)	**87.78(0.12)**	72.43(0.21)	**89.79(0.07)**
Grass_M	10	473	70.94(0.17)	**72.43(0.17)**	65.81(0.19)	**66.65(0.22)**	77.74(0.15)	**81.36(0.10)**	77.23(0.20)	**77.30(0.20)**
Grass_T	15	715	73.76(0.06)	**94.06(0.04)**	85.62(0.09)	**95.65(0.04)**	93.96(0.05)	**97.76(0.03)**	90.83(0.08)	**97.26(0.01)**
Grass_P	1	27	100.0(0.00)	**100.0(0.00)**	**98.80(0.03)**	71.20(0.23)	100.0(0.00)	**100.0(0.00)**	100.0(0.00)	**100.0(0.00)**
Hay_W	10	468	99.68(0.01)	**99.83(0.00)**	**95.87(0.05)**	89.85(0.07)	**99.91(0.11)**	92.85(0.13)	95.87(0.04)	**96.06(0.05)**
Oats	1	19	**72.63(0.13)**	61.58(0.15)	**46.31(0.16)**	22.11(0.14)	**91.57(0.11)**	83.16(0.12)	**78.94(0.30)**	72.11(0.37)
Soybean_N	20	952	77.88(0.05)	**85.80(0.04)**	69.56(0.09)	**82.17(0.05)**	86.62(0.05)	**87.38(0.05)**	78.17(0.06)	**83.32(0.06)**
Soybean_M	50	2405	83.53(0.05)	**87.65(0.05)**	82.01(0.05)	**85.36(0.04)**	89.33(0.15)	**91.99(0.15)**	86.63(0.04)	**91.37(0.03)**
Soybean_C	12	581	**56.03(0.11)**	54.46(0.11)	**57.05(0.17)**	49.54(0.13)	**69.88(0.18)**	68.82(0.18)	55.70(0.14)	**62.50(0.13)**
Wheat	4	201	91.13(0.05)	**99.36(0.01)**	83.94(0.12)	**97.68(0.04)**	98.76(0.04)	**99.80(0.04)**	93.10(0.06)	**99.36(0.01)**
Woods	26	1239	**90.09(0.06)**	87.53(0.06)	**91.70(0.05)**	88.58(0.04)	**94.63(0.14)**	92.98(0.14)	88.64(0.06)	**89.76(0.06)**
Buildings	8	378	**75.02(0.10)**	74.55(0.11)	75.73(0.18)	**77.80(0.11)**	77.48(0.12)	**77.49(0.15)**	72.19(0.17)	**74.50(0.17)**
Stone	2	91	94.89(0.05)	**98.80(0.00)**	61.74(0.22)	**65.43(0.43)**	74.24(0.09)	**98.04(0.06)**	97.61(0.02)	**98.69(0.01)**
OA			77.87(0.03)	**82.77(0.04)**	75.27(0.03)	**78.12(0.03)**	85.69(0.04)	**87.07(0.02)**	82.03(0.03)	**85.96(0.02)**
AA			77.73(0.03)	**82.67(0.03)**	70.63(0.02)	**71.35(0.05)**	86.03(0.04)	**88.21(0.03)**	83.04(0.04)	**86.81(0.04)**
Kappa			74.79(0.04)	**80.35(0.04)**	71.70(0.03)	**75.01(0.04)**	83.68(0.05)	**85.25(0.03)**	79.54(0.03)	**84.02(0.03)**

multilabels to single-labels, the number of other samples has increased, and the samples with small size are prone to be assigned to the classes with a large number of pixels. Furthermore, because the number of multilabel training samples is large and the label information provided by single-label samples is limited, the misclassified samples may play a major negative role.

Fig. 10 shows the classification maps yield by using different classifiers in the proposed method with single-label training samples or with both single-label and multilabel training samples along with the OAs. As Fig. 10 shows, it can be visually observed that all classifiers with both single-label and multilabel training samples yield better classification maps and provide higher OA reported. In addition, compared with the classifiers using only limited single-label training samples, the multilabel training samples can significantly improve the classification performance of several classes, i.e., Soybean_M and wheat. For more intuitive comparisons, histograms are added to compare the OAs of different classifiers with or without multi-label training samples. As Fig. 11(a) shows, the performance of all classifiers can be improved by using the multi-label training samples. Taking the SVM for example, compared with just using single-label samples for training, the proposed method using multi-label training samples increases the OA from 77.87% to 82.77%.

The second and third experiments are performed on the other two data sets, i.e., the Salinas image and the KSC image, respectively. The single-label training sets, which account for five labeled samples per class and three labeled samples per class, are chosen randomly. We use the average values of the experimental results repeated ten times as the ultimate results. As Tables IV and V and Fig. 11(b) and (c) show, it is clear to see that different classifiers with both single-label and multilabel training samples always obtain higher values in terms of the OA, AA, and Kappa.

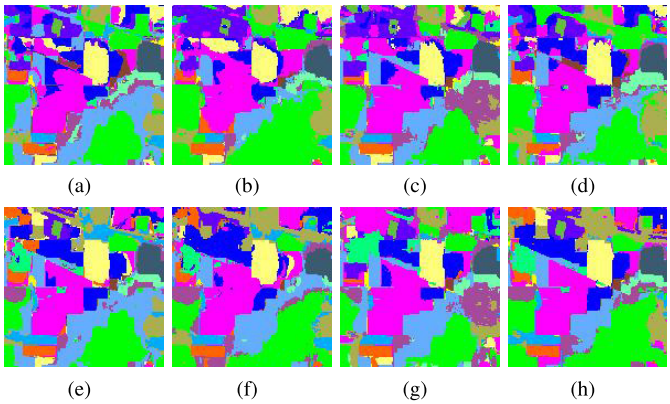


Fig. 10. Classification maps obtained by using different classifiers in the proposed method on the Indian Pines data set with or without multilabel training samples. (a) Classification maps obtained by NN with single-label training samples (NN_S). (b) RF_S. (c) LORSAL_S. (d) SVM_S. (e) Classification maps obtained by NN with both single-label and multilabel training samples (NN_M). (f) RF_M; (g) LORSAL_M; (h) SVM_M. (a) OA = 75.27%. (b) OA = 85.69%. (c) OA = 82.03%. (d) OA = 82.77%. (e) OA = 78.12%. (f) OA = 78.12%. (g) OA = 87.07%. *(h) OA = 85.96%.

For the LORSAL classifier, the OA has been significantly improved, but the LORSAL classifier with multilabel samples gets worse performance compared with single-label samples only for some classes. The reason is that the LORSAL algorithm based on multinomial logistic regression is sensitive with respect to the mislabeled samples. If the mislabeled samples are not completely removed and used for training, the final trained models may fit well with the mislabeled samples. The proposed spectral–spatial similarity-based mislabeled sample removal method is simple and basic, in order to solve this

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TABLE IV

OBJECTIVE ACCURACY INDEXES OBTAINED ON THE SALINAS DATA SET BY DIFFERENT CLASSIFIERS. "_S" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES, "_S+M" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES AND MULTILABEL TRAINING SAMPLES. THE STANDARD VARIANCES OF THE RESULTS OF TEN REPEATED EXPERIMENTS ARE PRESENTED IN THE PARENTHESES

Class	train	test	Classification accuracies obtained by the compared methods (in %)							
			NN_S	NN_S+M	RF_S	RF_S+M	LORSAL_S	LORSAL_S+M	SVM_S	SVM_S+M
Weeds_1	5	2004	96.13(0.06)	**98.63(0.02)**	93.28(0.07)	**98.93(0.01)**	99.25(0.01)	**100.0(0.00)**	96.12(0.07)	**100.0(0.00)**
Weeds_2	5	3721	74.90(0.14)	**85.54(0.03)**	**29.80(0.18)**	21.01(0.07)	89.37(0.10)	**96.37(0.01)**	66.36(0.17)	**76.14(0.01)**
Fallow	5	1971	99.90(0.00)	**99.90(0.00)**	99.99(0.00)	**100.0(0.00)**	98.88(0.02)	**100.0(0.00)**	100.0(0.00)	**100.0(0.00)**
Fallow_P	5	1389	97.59(0.03)	**97.59(0.03)**	**92.90(0.06)**	92.76(0.06)	97.97(0.02)	**98.09(0.02)**	98.01(0.03)	**100.0(0.00)**
Fallow_S	5	2673	**91.07(0.08)**	91.04(0.08)	82.14(0.12)	**86.51(0.09)**	**95.15(0.06)**	94.59(0.06)	92.82(0.07)	**94.46(0.07)**
Stubble	5	3954	99.59(0.00)	**99.59(0.00)**	99.61(0.00)	**99.68(0.00)**	**99.27(0.01)**	97.37(0.10)	99.75(0.00)	**99.80(0.00)**
Celery	5	3574	**99.64(0.00)**	99.49(0.00)	98.50(0.02)	**99.05(0.01)**	**98.91(0.02)**	97.75(0.03)	**99.89(0.00)**	99.75(0.00)
Grapes	5	11266	83.80(0.06)	**90.79(0.05)**	84.07(0.06)	**93.97(0.04)**	81.90(0.04)	**89.96(0.03)**	78.98(0.07)	**99.98(0.06)**
Soil	5	6198	99.99(0.00)	**99.99(0.00)**	99.62(0.01)	**99.63(0.01)**	**99.97(0.06)**	99.95(0.03)	99.98(0.00)	**99.98(0.00)**
Corn	5	3273	**89.90(0.11)**	89.78(0.11)	85.84(0.12)	**87.05(0.10)**	**95.69(0.08)**	87.82(0.08)	92.56(0.10)	**99.69(0.10)**
Lettuce_4	5	1063	**86.17(0.02)**	77.40(0.08)	**86.23(0.09)**	82.56(0.11)	81.15(0.04)	**81.84(0.05)**	**91.19(0.01)**	91.17(0.01)
Lettuce_5	5	1922	**98.03(0.01)**	92.09(0.10)	**96.51(0.04)**	71.58(0.11)	**99.00(0.00)**	98.67(0.11)	97.79(0.01)	**99.84(0.00)**
Lettuce_6	5	911	**81.38(0.01)**	80.33(0.01)	**75.87(0.04)**	67.86(0.05)	**89.43(0.05)**	83.50(0.07)	**85.62(0.05)**	83.31(0.01)
Lettuce_7	5	1065	97.29(0.02)	**97.29(0.02)**	**96.98(0.04)**	95.79(0.05)	**98.18(0.02)**	95.78(0.03)	96.36(0.03)	**99.81(0.03)**
Vinyard_U	5	7263	78.65(0.07)	**79.80(0.04)**	63.20(0.11)	**77.81(0.09)**	89.01(0.08)	**96.12(0.09)**	94.64(0.05)	**95.87(0.09)**
Vinyard_T	5	1802	**94.51(0.06)**	94.35(0.06)	95.84(0.02)	**98.80(0.00)**	**91.49(0.08)**	79.23(0.11)	97.95(0.02)	**99.06(0.02)**
OA			89.82(0.02)	**91.83(0.01)**	83.56(0.02)	**86.50(0.01)**	92.40(0.02)	**94.28(0.02)**	90.92(0.02)	**96.48(0.02)**
AA			91.79(0.01)	**92.10(0.01)**	86.27(0.02)	**85.81(0.01)**	91.11(0.01)	**93.57(0.02)**	93.00(0.01)	**96.03(0.01)**
Kappa			88.71(0.02)	**90.91(0.01)**	81.75(0.03)	**84.98(0.01)**	91.58(0.02)	**93.64(0.03)**	89.95(0.03)	**96.08(0.02)**

TABLE V

OBJECTIVE ACCURACY INDEXES OBTAINED ON THE KSC DATA SET BY DIFFERENT CLASSIFIERS. "_S" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES, "_S+M" REFERS TO CLASSIFIERS WITH SINGLE-LABEL TRAINING SAMPLES AND MULTILABEL TRAINING SAMPLES. THE STANDARD VARIANCES OF THE RESULTS OF TEN REPEATED EXPERIMENTS ARE PRESENTED IN THE PARENTHESES

Class	train	test	Classification accuracies obtained by the compared methods (in %)							
			NN_S	NN_S+M	RF_S	RF_S+M	LORSAL_S	LORSAL_S+M	SVM_S	SVM_S+M
Scrub	3	758	70.58(0.01)	**70.58(0.01)**	**72.68(0.01)**	72.41(0.02)	71.90(0.01)	**71.94(0.01)**	71.98(0.02)	**72.03(0.02)**
Willo_ S	3	240	95.83(0.18)	**99.17(0.18)**	**72.58(0.12)**	71.83(0.18)	80.46(0.19)	**84.88(0.18)**	75.58(0.18)	**76.54(0.18)**
Cabbag_P	3	253	99.60(0.01)	**99.60(0.01)**	**76.05(0.18)**	71.94(0.21)	98.26(0.05)	**98.81(0.03)**	61.46(0.27)	**63.36(0.26)**
Cabbage_O	3	249	82.73(0.14)	**84.34(0.20)**	88.43(0.13)	**95.38(0.12)**	76.87(0.14)	**96.75(0.07)**	94.50(0.06)	**98.51(0.05)**
Slash_P	3	158	98.73(0.08)	**98.73(0.06)**	**87.91(0.08)**	84.75(0.07)	**91.71(0.09)**	85.32(0.08)	**85.32(0.08)**	84.43(0.06)
Oak_H	3	226	55.31(0.24)	**55.31(0.24)**	95.53(0.14)	**97.35(0.08)**	**77.52(0.24)**	74.12(0.23)	**83.14(0.22)**	82.12(0.23)
Hardwood_S	3	102	100.0(0.11)	**100.0(0.11)**	100.0(0.00)	**100.0(0.00)**	96.27(0.07)	**96.67(0.07)**	100.0(0.00)	**100.0(0.00)**
Graminoid_M	3	428	**89.02(0.18)**	88.55(0.16)	75.56(0.19)	**80.77(0.17)**	**99.86(0.00)**	93.18(0.08)	**85.51(0.19)**	85.09(0.19)
Spartina_M	3	517	38.68(0.21)	**38.68(0.21)**	56.50(0.21)	**80.14(0.07)**	**66.83(0.19)**	60.27(0.18)	50.12(0.10)	**73.66(0.07)**
Cattail_M	3	401	**35.41(0.15)**	33.67(0.07)	**39.23(0.07)**	36.56(0.02)	**56.01(0.18)**	41.72(0.10)	48.18(0.08)	**57.36(0.06)**
Salt_M	3	416	100.0(0.04)	**100.0(0.19)**	**91.20(0.18)**	74.64(0.23)	**96.03(0.09)**	90.10(0.19)	86.59(0.22)	**86.59(0.22)**
Mud_F	3	500	86.80(0.00)	**87.40(0.02)**	86.92(0.05)	**87.40(0.01)**	85.18(0.03)	**86.98(0.01)**	72.40(0.09)	**86.30(0.01)**
Water	3	924	100.0(0.00)	**100.0(0.00)**	85.55(0.17)	**92.35(0.10)**	54.86(0.18)	**100.0(0.00)**	96.32(0.08)	**99.71(0.01)**
OA			79.33(0.03)	**80.45(0.02)**	76.79(0.04)	**79.35(0.02)**	75.93(0.05)	**82.21(0.03)**	76.82(0.03)	**82.06(0.02)**
AA			80.98(0.03)	**81.23(0.03)**	79.09(0.03)	**80.42(0.02)**	80.94(0.04)	**83.10(0.03)**	77.78(0.03)	**81.98(0.02)**
Kappa			77.13(0.03)	**77.28(0.03)**	74.32(0.04)	**77.14(0.02)**	73.60(0.05)	**80.30(0.03)**	74.36(0.03)	**80.12(0.02)**

problem, combining it with more precise mislabeled training samples detection and correction methods to achieve optimization of multilabel training sets will be our future work. Furthermore, using the classifiers that are robust to the wrong labels in the training set is also a viable solution. For example, the SVM classifier can effectively utilize the multilabel training samples because it is able to screen the training samples by introducing the slack variable, and thus, it is more robust to the mislabeled samples in the classified multilabel training set. For the KSC image, compared to the SVM classifier trained with limited single-label training samples, the OA can be increased from 76.82% to 82.06% by using the proposed multilabel sample augmentation method; for the Salinas image, the value of OA is significantly increased from 90.92% to 96.48%.

Compared to data sets with an imbalanced sample size, the proposed method is more suitable for the data sets with a balanced sample size. For the multilabel sample annotation, the pixels in the ground truths are labeled by giving multiple labels that all classes contained in the pixel neighborhood. If the number of training samples belonging to certain classes is too small or they are scattered distributed, after the conversion from multilabels to single-labels, these samples may be misclassified to the classes whose numbers are large and removed. The small classes may be compromised by the

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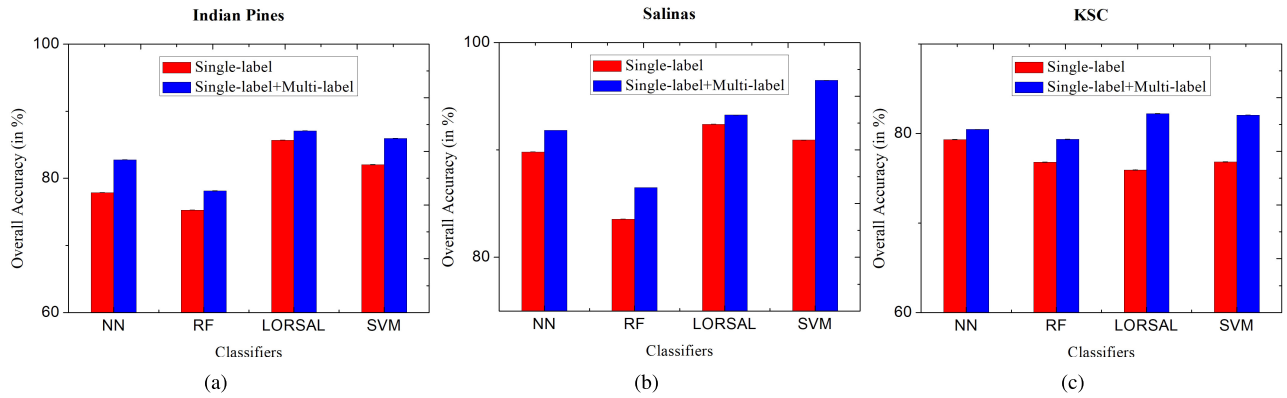


Fig. 11. Performance comparison (OAs) of different classifiers with or without multilabel training samples. (a) Indian Pines data set. (b) Salinas data set. (c) KSC data set.

large classes, which may lead to a reduction in classification accuracy. The Indian Pines data set consists of imbalanced classes of various sizes. There are four classes of objects with less than 100, including class 1 with 46 labeled samples, class 7 with 28 labeled samples, class 9 with 20 labeled samples, and class 16 with 93 labeled samples. There are three classes of objects with more than 1000, including class 2 with 1428 labeled samples, class 11 with 2455 labeled samples, and class 14 with 1265 labeled samples. Therefore, the Indian Pines data set can be used to demonstrate the effectiveness of the proposed method when it is used on data sets with imbalanced sample size. As Table III shows, for all classifiers used in the experiments, the classification performance of most classes can be improved by using the multilabel training samples. Especially, for classes 1, 7, and 9 with small numbers of training samples, the classification accuracies can be improved by using the multilabel training samples in most cases. Taking the SVM classifier for example, for class 1, the proposed method with multilabel training samples can increase the OA from 89.11% to 97.78%.

TABLE VI

OAs, AAs, and Kappa Coefficients of the Proposed Method With Different Multilabel Training Samples. "Number" Refer to the Multilabel Training Samples in the ith Multilabel Region and "Labels" Refer to the Labels of Multilabel Training Samples

Data Sets	Number	Labels	OA	AA	KAPPA
Indian Pines	None	None	82.04	83.04	79.55
	1-th	2-6;10;12;15;16	82.93	84.63	80.61
	2-th	2;10;11	82.65	83.41	80.25
	3-th	1;2;6;10;11	83.04	84.11	80.69
	4-th	3;5;11;13	83.21	83.77	80.84
	ALL	ALL	85.96	86.81	84.01
Salinas	None	None	90.92	93.00	89.95
	1-th	2;8	93.02	93.96	92.26
	2-th	8;15	93.32	93.59	92.58
	3-th	1;11-13	91.22	93.66	90.28
	ALL	ALL	96.48	96.03	96.08
KSC	None	None	76.82	77.78	74.36
	1-th	12;13	78.86	79.22	76.57
	2-th	2-5;12	79.97	80.17	77.82
	ALL	ALL	82.06	81.98	80.12

E. Effect of Different Multilabel Training Samples on Classification Performance

The objective of this section is to evaluate the generalization capabilities of the proposed multilabel sample augmentation method with different numbers of multilabel training samples. In the first experiment, we evaluate the performance of the proposed method using different multilabel regions as training samples. This is important since it is directly related to the labeling way of multilabel pixels in hyperspectral images for classification. In this experiment, the single-label samples are selected ten times randomly and the numbers of them are the same as those shown in Tables III–V. We use the average values of the experimental results repeated as the ultimate results for a fair comparison. The experimental results on three hyperspectral images are shown in Table VI. In this table, "Number" represents that the ith multilabel region constructed earlier is used for classification [see Figs. 5(c)–7(c)] and "Labels" denotes the multiple labels of multilabel samples. "None" represents that just the single-label samples are used for training, "ALL" represents that both the single-label samples and all multilabel training samples are used.

It can be seen from Table VI that the multilabel training samples have different levels of improvement on classification performance in terms of OA, AA, and Kappa with the same single-label samples. This experiment result demonstrates that annotating training examples in a multilabel way can contribute to the improvement of classification accuracy. For the 2th multilabel region of the KSC data set, 327 pixels are labeled as 12 and 13. The OA is improved from 76.82% to 78.86%. Compared to classification with a large number of single-label training samples, the employment of multilabel training samples can not only reduce the manpower required but also improve the classification accuracies.

Furthermore, we evaluate the influence of the number of multilabel training samples on classification performance as a function of the number of multilabel training samples. Different numbers of multilabel samples are selected randomly from the multilabel samples constructed before [see Figs. 5(c)–7(c)]. We consider a varying number of multilabel

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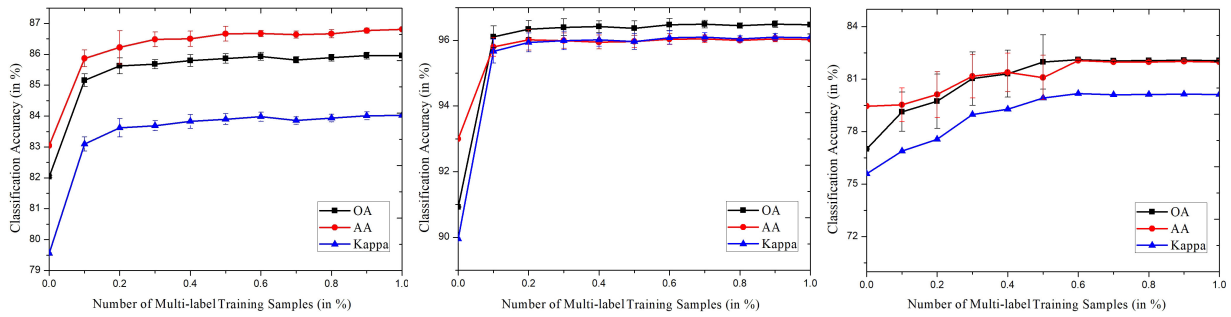


Fig. 12. Performance comparisons of the SVM classifier trained using the limited single-label and different numbers of multilabel training samples. Experiments are performed on three hyperspectral data sets with different numbers of multilabel training samples (vary from 10% to 100% for each multilabel region). (a) OAs, AAs, and Kappas recorded on the Indian Pines test image. (b) Salinas test image. (c) KSC test image.

TABLE VII

NUMBER OF SINGLE-LABEL TRAINING SAMPLES REQUIRED FOR THE RF AND SVM CLASSIFIERS TO ACHIEVE THE OBJECTIVE OAS ON THE INDIAN PINES AND SALINAS DATA SETS. 'S" REFERS TO CLASSIFICATION WITH SINGLE-LABEL TRAINING SAMPLES, "S+M" REFERS TO CLASSIFICATION WITH SINGLE-LABEL AND MULTILABEL TRAINING SAMPLES (THE PROPOSED MULTILABEL SAMPLE AUGMENTATION METHOD). THE STANDARD VARIANCES OF THE TEN REPEATED EXPERIMENTAL RESULTS ARE PRESENTED IN THE PARENTHESES

Data Set	Indian Pines				Salinas			
Method	RF_S	RF_S+M	SVM_S	SVM_S+M	RF_S	RF_S+M	SVM_S	SVM_S+M
Number	165	**111**	165	**111**	112	**80**	160	**80**
OA	82.07(0.01)	**82.34(0.03)**	89.09(0.01)	**89.32(0.02)**	86.37(0.02)	**87.23(0.01)**	96.05(0.01)	**96.48(0.02)**
AA	**75.75(0.01)**	69.23(0.05)	88.31(0.02)	**89.73(0.04)**	**88.67(0.01)**	86.43(0.01)	**97.24(0.01)**	96.48(0.01)
Kappa	79.48(0.01)	**79.78(0.04)**	87.53(0.01)	**89.73(0.03)**	84.86(0.02)	**85.79(0.01)**	95.62(0.01)	**96.48(0.02)**

training samples ranging from 10% to 100% of the multilabel ground truth, averaged over ten realizations. The multilabel training samples of the same size are selected ten times and each of them is evaluated by using the proposed method with randomly selected single-label training samples. These multilabel training samples are selected from different multilabel regions according to the predetermined proportion. We select the SVM classifier as the base-level single-label classifier, while individual parameters of the SVM classifier are instantiated according to [28]. In order to avoid randomness of the constructed multilabel training samples, the average values of ten times repeated experimental results are used as the final results.

In Fig. 12, we present the variation of tendencies of the values of the OA, AA, and Kappa coefficient on three hyperspectral images as the number of multilabel training samples increases. One can observe that, for all metrics, the performance increases monotonically with the gradual increase of the multilabel training set size in most cases, with a fast rate during initial cases and then with a slower one. The result indicates that the increase in the number of multilabel training samples has a beneficial effect on the performance of the proposed training sample augmentation method. However, the multilabel training set size plays a less role in the improvement of classification performance after reaching a certain accuracy. The reason is that when the number of multilabel training samples is large and the exact label information provided by single-label samples is limited, the misclassified samples may play a negative role. In order to solve the problem, the multilabel classification problem is expected to be combined with more precise mislabeled training samples detection and correction methods to achieve optimization of the training set.

F. Computation Complexity of the Proposed Method

In order to demonstrate that the multilabel samples can reduce the number of single-label samples required, an experiment is performed on the Indian Pines and Salinas data sets. The RF and SVM classifiers and multilabel training samples mentioned before are still used. In the experiment, besides the same single-label samples used for training at first, the new added single-label training samples are selected randomly by increasing the same number of samples for each class. In this situation, in order to do a fair comparison, besides comparing the classification accuracy, Table VII compares the numbers of single-label training samples required for different classifiers to achieve approximately equal OA values. It can be observed that the proposed method requires fewer samples in all cases. For example, for the Salinas data set, with 80 single-label samples and the multilabel samples, the SVM classifier is able to achieve an OA of 96.48%. However, the SVM classifier requires more than 160 samples to achieve an approximate accuracy, which means that the three multilabel regions can replace 80 single-label training samples. This experiment further demonstrates that the proposed method is able to make full use of the multilabel samples, thus, able to dramatically reduce human efforts in labeling accurate single-label training samples.

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TABLE VIII
COMPUTATION TIMES (SECONDS) OF THE ALGORITHMS FOR THE EXPERIMENTAL DATA SETS

Time (s)	NN	RF	LORSAL	SVM
Indian Pines	5.29	20.62	6.15	5.91
Salinas	12.29	44.71	12.68	12.65
KSC	20.78	92.23	21.28	21.53

It should be mentioned that the added single-label training samples are randomly selected from the hyperspectral data, which is a natural choice, since all labeled data are treated equally and each sample would be selected with the same probability. The labeled samples are first grouped into subsets based on their labels, and then, random sampling is carried out within each subset. The number of newly added training samples belonging to each class is balanced. However, the constructed multilabel samples do not contain all classes and contain different numbers of samples belonging to different classes. This is why the values of AA obtained by using the single-label training samples are higher in some cases. If the added single-label training samples are selected from the multilabel regions, more single-label training samples would be required to achieve an approximate OA.

Furthermore, the computation complexity of the recursive filtering operation is $O(N)$, and thus, the IFRF method has a computational complexity of $O(KN)$, where N is the number of pixels and K is the number of features. The complexity of the ERS algorithm approximates $O(|V|\log|V|)$, where V is the vertex set. The complexity of the spectral–spatial similarity-based mislabeled sample removal approximates $O(|D||S| + |D|C\log C)$, where $|D|$ is the number of multilabel samples, $|S|$ is the number of single-label samples, and C is the number of classes. In our experiments, the computation cost of the proposed methods is measured on a laptop with 2.80 GHz CPU and 4 GB memory. The experiment is implemented in MATLAB. The computing times of the proposed method using four classifiers on three experimental data sets are given in Table VIII. As this table shows, the proposed method is efficient.

V. CONCLUSION

This article aims at presenting a radically different training sample augmentation approach for hyperspectral image classification by extending and developing the concept of multilabel for the first time. Unlike the multilabel problem in computer vision, the proposed multilabel sample annotation method can reduce the pressure of human annotation by labeling pixels in certain regions simultaneously instead of labeling the pixels one by one. We can annotate multilabel samples easily and use them as weak label information for hyperspectral image classification.

We also propose a sample augmentation method to utilize both the single-label and multilabel training samples. This multilabel sample augmentation method is successfully applied for hyperspectral image classification. By performing experiments on three real hyperspectral data sets, it can be demonstrated that the performance of several classical classifiers can be improved remarkably by using the proposed multilabel sample augmentation method. It is found that very easy methods can realize the improvement of classification performance by using the multilabel training samples, which demonstrates that the multilabel training sample augmentation method is promising in real applications.

VI. OUTLOOKS

In this article, we reviewed the current state of the art in hyperspectral image classification with single-label and multilabel training samples. Remote sensing researchers have exploited many powerful hyperspectral image classification methods, and some promising results have been achieved. One of the most potential problems to be further explored is multilabel hyperspectral image classification, which is still relatively young in the field. In the upcoming years, the rapid advancement of multilabel problems in remote sensing is expected. Here, by analyzing the limitations of the proposed method, some outlooks are given:

1) In the proposed method, a simple spectral–spatial similarity-based mislabeled sample detection method is used to remove the mislabeled samples. The reason is that when the multilabels of samples are converted into single labels, mislabeled samples are inevitably generated. When the number of generated mislabeled samples is too large, the ability of the spectral–spatial similarity-based method to remove mislabeled samples is limited, and the mislabeled samples that are not removed may have a negative impact on the classifiers. Furthermore, the proposed method only detects the mislabeled samples and does not correct them. If the mislabeled samples can be effectively corrected based on the label correlation information, spatial and spectral characteristics of the samples, the accuracy will be expected to be further improved. In order to solve these problems, combining the multilabel classification problem with more precise mislabeled training samples detection and correction methods to achieve optimization of multilabel training set will be our future work. For example, the difference between the same class and the similarity between different classes in the training set will be taken into account. Particularly, most of the hyperspectral mislabeled training samples detection and correction methods are proposed for incorrect labels that are randomly generated. For multilabel problems, the incorrect labels are limited to the classes to which the multilabel belongs (spatially adjacent classes), which is a bigger challenge for detection and correction.
2) In the field of hyperspectral classification, there is no multilabel data set that can be directly used, so we built simulation data based on three common data sets for experiments. In the future work, combined with the application requirements in the real scenario, according to the annotation difficulty and contribution of multilabel training samples in different applications, studying the fast and efficient single-label and multilabel sample training set labeling scheme is very meaningful. For example, active learning, which can effectively reduce

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unnecessary labeling efforts, is also expected to be employed. Furthermore, besides the block-based multilabel annotation, we can design spatial and spectral characteristics-based adaptive multilabel annotation algorithms. Based on this, the hyperspectral image classification algorithms for different complex scenes under the supervision of both single-label and multilabel samples can be explored.

3) How to make full use of the weak label information provided by the multilabel samples is the key to the problem. The weak supervised learning and semisupervised learning, which have been widely employed to solve hyperspectral classification problems with a small-size sample, are expected to be introduced in the future research. For example, combining the proposed method with semisupervised learning, which unavoidably produce mislabeled samples, will be a promising research direction. Through integrating the proposed method with weak supervised learning or semisupervised learning, the robustness of the proposed method is also expected to be further improved.
4) If the number of training samples belonging to certain classes is too small or they are scattered distributed, the classes with small size may be compromised by classes with large size, which may lead to a reduction in classification accuracy. For the data sets with an imbalanced sample size, after the conversion from multilabels to single-labels, the samples with small sizes are prone to be assigned to the classes with a large number of pixels. For such an imbalanced problem, using the classifiers designed for imbalanced data sets is a viable strategy.

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